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Assignment 3: Office network communication

https://github.com/elovric13/eutech.git

1. Install VirtualBox and set up a small office simulation

Four virtual machines were created in VMware Workstation to simulate the office environment: one Ubuntu Server 22.04 (acting as the central server) and three Ubuntu Desktop 22.04 client machines. All VMs were connected to the same internal virtual network to allow communication between them.

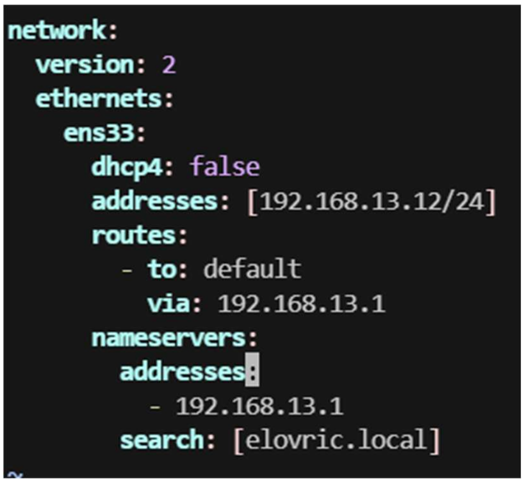
2. Assign static IP addresses and configure subnetting

A 192.168.13.0/24 subnet was chosen for the office network. Static IP addresses were assigned to machines.

Host	IP Address	Subnet Mask	Gateway
Ubuntu26 (Server)	192.168.13.1	255.255.255.0	192.168.13.1
PC1	192.168.13.11	255.255.255.0	192.168.13.1
PC2	192.168.13.12	255.255.255.0	192.168.13.1
PC3	192.168.13.13	255.255.255.0	192.168.13.1

```
network:
  version: 2
  ethernets:
    ens33:
      dhcp4: true
    ens37:
      dhcp4: false
      addresses:
        - 192.168.13.1/24
```

Figure 1: Server network configuration



```
network:
  version: 2
  ethernets:
    ens33:
      dhcp4: false
      addresses: [192.168.13.12/24]
      routes:
        - to: default
          via: 192.168.13.1
      nameservers:
        addresses:
          - 192.168.13.1
        search: [elovric.local]
```

Figure 2: Client network configuration

```
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gro
up default qlen 1000
    link/ether 00:0c:29:9a:ba:2f brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.13.12/24 brd 192.168.13.255 scope global noprefixroute ens33
       valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe9a:ba2f/64 scope link
       valid_lft forever preferred_lft forever
```

Figure 3: Client ipconfig

3. Create a network topology diagram

A logical network topology diagram was created using draw.io, illustrating the connections between the server, the three clients, and the shared printer within the configured subnet.

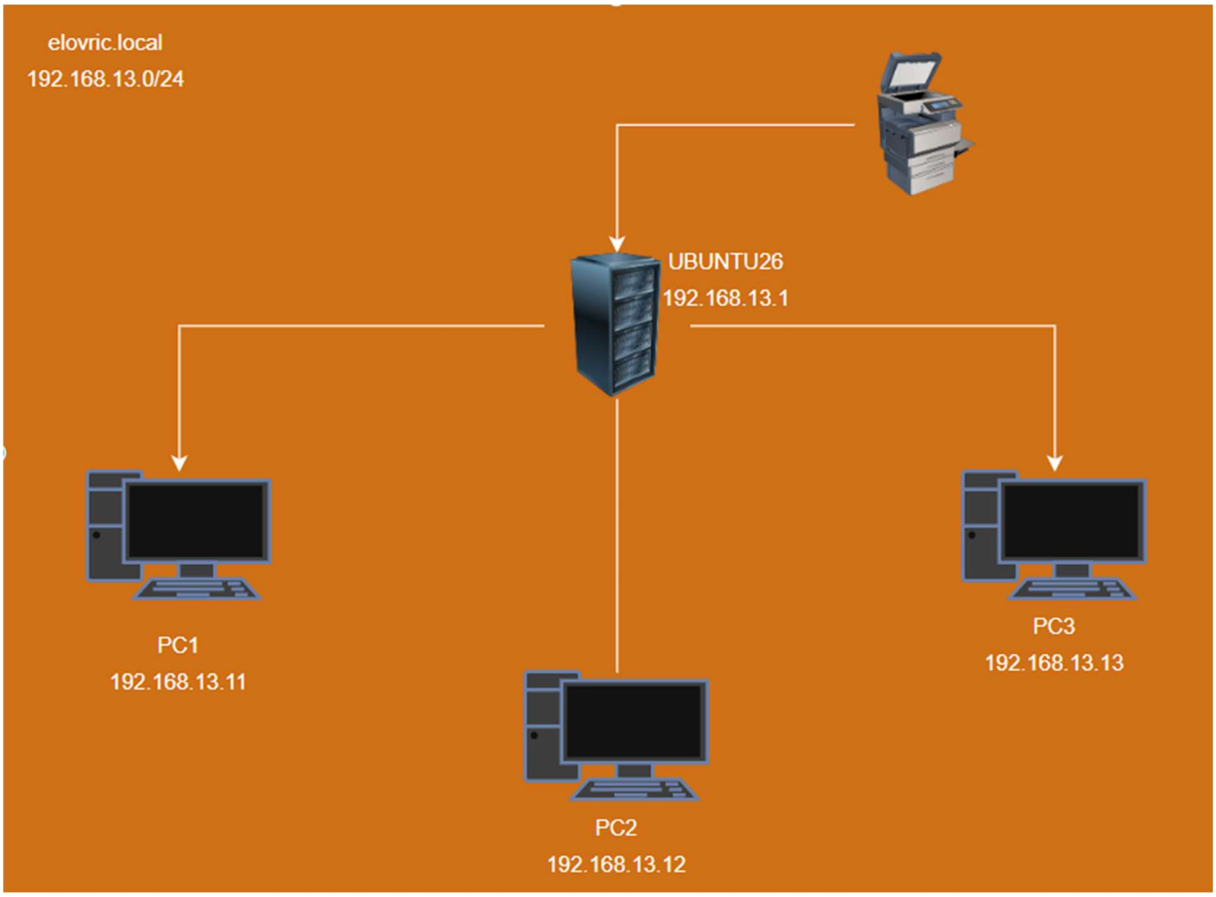


Figure 4: Network topology

4. Set up local DNS

Local DNS resolution was configured using dnsmasq on the server. Dnsmasq was installed and configured to act as a lightweight DNS server for the local network. The client machines were then configured to use the server's IP as their DNS server.

```
listen-address=192.168.13.1
bind-interfaces
domain=elovric.local
address=/ubuntu26.elovric.local/192.168.13.1
address=/pc1.elovric.local/192.168.13.11
address=/pc2.elovric.local/192.168.13.12
address=/pc3.elovric.local/192.168.13.13
"/etc/dnsmasq.conf" [readonly] 697L, 28120B
```

Figure 5: dnsmasq configuration

```
pc2@pc2:~$ ping pc1.elovric.local
PING pc1.elovric.local (192.168.13.11) 56(84) bytes of data.
64 bytes from pc1.elovric.local (192.168.13.11): icmp_seq=1 ttl=64 time=1.30 ms
64 bytes from pc1.elovric.local (192.168.13.11): icmp_seq=2 ttl=64 time=1.19 ms
64 bytes from pc1.elovric.local (192.168.13.11): icmp_seq=3 ttl=64 time=2.02 ms
```

Figure 6: Successful ping and name resolution

```
pc2@pc2:~$ nslookup pc3.elovric.local
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   pc3.elovric.local
Address: 192.168.13.13
```

Figure 7: Successful name resolution

5. Configure shared folders and printer

Samba was installed on the server for share network folder access. CUPS was configured on the server to share a virtual PDF printer.

Figure 8: Shared file access from a client

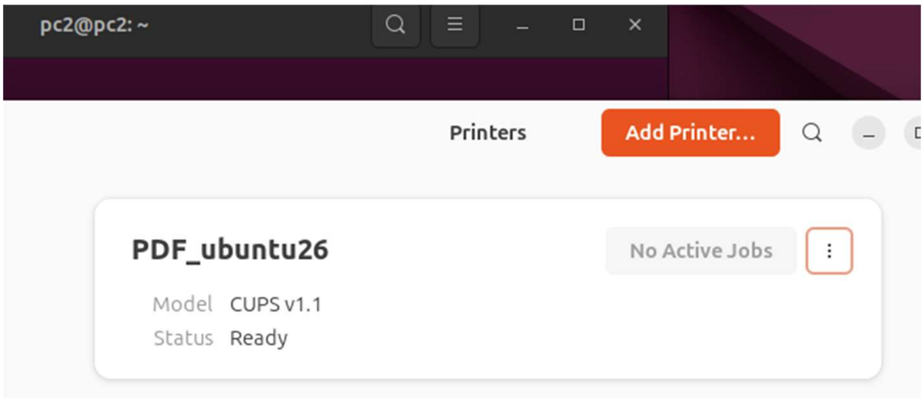


Figure 9: Printer access from a client

6. Implement automatic daily backups

A backup was implemented using rsync. A cron job was configured on the client machines to automatically synchronize ~/Documents folder to a backup directory on the server every day at a 2AM.

```bash
#!/bin/bash

CLIENT_NAME="pc2"
SOURCE_DIR="/home/$USER/Documents/"
SERVER_USER="backupuser"
SERVER_HOST=192.168.13.1
DEST_DIR="/backups/$CLIENT_NAME"
SSH_KEY="/home/$USER/.ssh/backup_key"

DATE=$(date +"%Y-%m-%d_%H-%M-%S")
BACKUP_DEST="$DEST_DIR/$DATE"

rsync -avz --delete \
  -e "ssh -i $SSH_KEY -o StrictHostKeyChecking=no" \
  "$SOURCE_DIR" \
  "$SERVER_USER@$SERVER_HOST:$BACKUP_DEST" \
  2>&1

if [ $? -eq 0 ]; then
    echo "[$DATE] Backup SUCCESSFUL."
else
    echo "[$DATE] Backup FAILED."
    exit 1
fi
```

Figure 10: rsync backup script

```
0 2 * * * /usr/local/bin/backup.sh
```

Figure 11: Configured cronjob

7. Simulate failure and recovery

To test the backup, a file was deleted from a client machine. The file was then successfully restored by copying it back from the server's rsync backup directory.

```
pc2@pc2:~$ cat Documents/test_backup_restore.txt
This is the file for testing backup and restore.

pc2@pc2:~$ /usr/local/bin/backup.sh
sending incremental file list
created directory /backups/pc2/2026-04-22_01-17-35
./
test_backup_restore.txt

sent 200 bytes  received 93 bytes  53.27 bytes/sec
total size is 50  speedup is 0.17
[2026-04-22_01-17-35] Backup SUCCESSFUL.
pc2@pc2:~$
pc2@pc2:~$ rm Documents/test_backup_restore.txt
```

Figure 12: Running the script and file deletion

```
pc2@pc2:~$ rsync -avz -e "ssh -i ~/.ssh/backup_key" \
> backupuser@192.168.13.1:/backups/pc2/2026-04-22_01-17-35 ~/Documents
receiving incremental file list
2026-04-22_01-17-35/
2026-04-22_01-17-35/test_backup_restore.txt

sent 47 bytes  received 239 bytes  63.56 bytes/sec
total size is 50  speedup is 0.17
pc2@pc2:~$
pc2@pc2:~$ cat Documents/2026-04-22_01-17-35/test_backup_restore.txt
This is the file for testing backup and restore.
```

Figure 13: rsync restore from server

```
pc2@pc2:~$ cat /var/log/backup_pc2.log
sending incremental file list
created directory /backups/pc2/2026-04-22_01-14-10
./
2026-04-22_00-09-47/
2026-04-22_00-09-47/test_backup_file.txt
2026-04-22_00-14-09/
2026-04-22_00-14-09/test_backup_file.txt
2026-04-22_00-14-09/2026-04-22_00-09-47/
2026-04-22_00-14-09/2026-04-22_00-09-47/test_backup_file.txt

sent 504 bytes  received 147 bytes  118.36 bytes/sec
total size is 49  speedup is 0.08
sending incremental file list
created directory /backups/pc2/2026-04-22_01-17-35
./
test_backup_restore.txt

sent 200 bytes  received 93 bytes  53.27 bytes/sec
total size is 50  speedup is 0.17
```

Figure 14: Backup logs